

Innovation at the intersection of energy, AI, and community.

How tying the AI infrastructure buildout to community ownership of renewable energy can resolve the standoff — and revive the towns that host it.

01 · The Buildout

A twenty-year compute migration, compressed into an investment cycle.

The current wave of data-center construction is better understood as the acceleration of a twenty-year trend than as a sudden event. U.S. data-center electricity consumption tripled between 2014 and 2023, rising from 58 TWh to 176 TWh — about 4.4% of national electricity use — as computing migrated from on-premise servers to hyperscale cloud facilities. The AI cycle has compressed the next leg of that curve into a few years: the Department of Energy projects data centers will consume between 6.7% and 12% of U.S. electricity by 2028.

The capital commitments behind that curve are without modern precedent. The four largest hyperscalers — Amazon, Microsoft, Alphabet, and Meta — have guided to combined 2026 capital expenditures approaching \$700 billion, the large majority directed at data centers, power, and networking. Independent estimates put AI data centers, hardware, and networking at roughly 1.4% of U.S. GDP in early 2026 — double the share of a year earlier — with full-year investment projected near 2% of GDP, a scale comparable to peak years of the railroad and telecom buildouts.

What has changed is the binding constraint. Chips and capital are available; sited land with power, water, transmission access, and local permission is not. Interconnection queues now hold roughly 2.6 TW of proposed generation and storage — about twice the installed capacity of the entire U.S. grid — with typical waits measured in years. In that environment, the scarce asset is not the data center itself but a permitted site with energy attached. That is the asset Community Compute is organized to produce, and it is produced only with community consent.

SOURCES

U.S. DOE / Lawrence Berkeley National Laboratory, 2024 Report on U.S. Data Center Energy Use. CNBC (Feb 2026), tech AI spending. Epoch AI, data-center share of GDP; Forbes (Jun 2026). LBNL, Queued Up 2024 / Utility Dive, 2.6 TW in interconnection queues.

02 · The Backlash

Opposition as a price signal.

Local resistance to energy and industrial siting is not new — wind, transmission, and pipeline projects have faced organized opposition for decades. What distinguishes the data-center backlash is its breadth. Gallup's first national survey on the question, fielded in March 2026, found 70% of Americans opposed to AI data-center construction in their local area, with 48% strongly opposed;

majorities of Democrats, Republicans, and independents all opposed local projects. Among elected officials who have publicly opposed projects, the split is 55% Republican to 45% Democratic — a distribution almost no other live issue in American politics produces.

The economic grievances behind those numbers are concrete rather than symbolic. Residential electricity prices nationally have risen roughly 42% over five years, and considerably faster in constrained regions. In PJM, the nation's largest wholesale market, the independent market monitor attributed a 174% increase in capacity costs for the 2025–26 delivery year — about \$9.3 billion — to data-center demand growth.

Read as a price signal rather than an obstacle, the opposition is informative: it says the prevailing development model socializes costs — rates, water, land, and construction disruption — while concentrating benefits in a small number of construction-phase jobs and negotiated tax payments that communities discount heavily. Data Center Watch tallies \$64 billion in projects blocked or delayed by local opposition over two years. That asymmetry between who pays and who collects is the specific problem Community Compute's model is designed to correct.

SOURCES

Gallup (Mar 2026), Americans Oppose AI Data Centers in Their Area. Data Center Watch, blocked/delayed report and partisan split. U.S. EIA residential price data via PolitiFact (Jun 2026). PJM Independent Market Monitor 2025 report.

03 • The Freeze

From incentive competition to structural risk.

For two decades, states competed for data centers with sales-tax exemptions and expedited permitting; more than thirty states adopted incentive programs. The reversal now underway is abrupt by the standards of state policy. In the first six weeks of the 2026 sessions, legislators filed more than 300 data-center-related bills across more than 30 states — including moratoriums, incentive rollbacks in Virginia, Georgia, and Oklahoma, and, in at least 18 states, special rate classes intended to insulate residential customers from large-load costs.

The executive actions have been more consequential than the bills. New York's statewide moratorium, announced July 14, 2026, paused permits for new hyperscale facilities for up to a year and halted an estimated \$10 billion of development. Three weeks later, Texas — the most active data-center market in the country — froze new grid-connection approvals pending an audit of the more than 1,800 projects in ERCOT's interconnection queue, which together request 474 GW, roughly five times the system's historical peak. When the two states most identified with opposite

regulatory philosophies converge on the same instrument within a month, developers should treat the change as structural rather than cyclical.

For project underwriting, the consequence is that regulatory and community risk is now the largest variable in time-to-power — itself the variable hyperscalers price most heavily. A development approach that resolves community terms *before* permitting, rather than defending a completed design against opposition, converts an unpriceable political risk into a known, contractable cost. That is the underwriting logic of the Community Compute model: benefit terms negotiated first, permits sought with the community as counterparty and co-beneficiary rather than adversary.

SOURCES

MultiState (Feb 2026), State Data Center Legislation in 2026. Gov. Hochul press release (Jul 14, 2026); Bisnow. Texas Tribune (Aug 3, 2026), ERCOT audit freeze.

04 • The Other Crisis

Falling populations and consolidating farms, where the buildout wants to go.

The demographic deterioration of rural America is a long secular trend, not a recent development. Roughly 40% of nonmetro counties reached their peak population in 1950 or earlier, and the 2010s were the first decade on record in which nonmetro America as a whole lost population between censuses; about two-thirds of nonmetro counties shrank over that decade. The drivers have shifted from out-migration alone to natural decrease: in the 2010s, 55% of nonmetro counties recorded more deaths than births, up from 22% in the 1990s.

Agricultural consolidation is following the same curve. The 2022 Census of Agriculture recorded 141,733 fewer farms than in 2017 — a 7% decline in five years — and 20.1 million fewer acres of farmland, an area the size of Maine. The remaining production is concentrating rapidly: farms grossing over \$1 million now account for 79% of agricultural product value, up from 69% five years earlier. For the operators who remain, the economics are thin; average cash rent for non-irrigated cropland runs \$146 per acre nationally.

These counties nonetheless hold precisely the assets the AI buildout requires: large contiguous parcels, existing transmission corridors, water in many basins, and county governments capable of moving quickly. The fiscal mathematics are favorable in both directions — a single facility can dominate a rural county's tax base, and the incremental revenue lands where school districts and services are otherwise contracting. The coincidence of falling land values and populations in exactly

the geography where land-intensive infrastructure demand is arriving is the arbitrage at the center of this paper.

SOURCES

Johnson & Lichter, RSF Journal (2025), rural depopulation. American Farm Bureau Federation, 2022 Census of Agriculture analysis. USDA cash-rent data via AFBF.

05 · The Thesis

Ownership, not compensation — the cooperative precedent.

There is an American precedent for solving exactly this shape of problem. In 1935, roughly nine in ten rural households lacked electric service because the economics of extension did not pencil for private utilities; member-owned rural electric cooperatives, capitalized with federal credit, electrified most of the countryside within two decades, and roughly 900 cooperatives still serve about 42 million people across more than half the country's land area. The lesson is institutional rather than nostalgic: rural infrastructure gets built at scale when the ownership structure gives rural residents a durable claim on the asset's cash flows.

The community's permission should be exchanged not for one-time payments, but for equity participation in the renewable generation, storage, and infrastructure that serves the data center.

Ownership changes the arithmetic on both sides. For residents, a project that pays lease income, dividends, and taxes for thirty years is categorically different from one that pays impact fees once. For developers, a community with a continuing financial stake becomes a partner in uptime, expansion, and future permitting rather than a permanent litigation risk.

Nor does the model depend on developer generosity. Utility-scale solar is the cheapest new generation available in most of the country, with levelized costs of \$38–78 per MWh before subsidies, and land-lease offers to farmers already run \$1,000–1,500 per acre annually against \$146 average cropland rents. There is ample margin in these projects to fund meaningful community ownership without impairing returns; what has been missing is a developer whose model routes that margin locally by design.

SOURCES

06 • The Mechanism

Making a market between hyperscalers and host communities.

The transaction at the core of the model is a straightforward exchange of the two things each side values most. Communities control permitting, land assembly, and political consent — the inputs that now determine whether a project happens at all. Hyperscalers and their development partners control capital and long-term offtake. Community Compute structures the trade: the community grants siting and permits on negotiated terms, and in exchange receives a defined package that can include equity in generation and storage assets, discounted or stabilized power, water and grid-infrastructure investment, and payments tied to project performance rather than one-time approval.

COMMUNITIES GET

Ownership stakes, stable revenue, lower energy costs, and infrastructure upgrades.

DEVELOPERS GET

Permits, social license, faster timelines, and durable local partnerships.

Each element of that package has an existing market analogue, which matters for both credibility and underwriting. Solar land leases at \$1,000–1,500 per acre are already six to ten times typical crop rents. Community-benefit agreements have moved from the margins to standard practice. Microsoft has committed to full recovery of the electricity costs its facilities impose on host communities, and the major hyperscalers jointly pledged at the White House to fund power and grid upgrades associated with their projects. The components are all in use somewhere; what does not yet exist is a standard package, priced and replicable across counties.

That absence is the market gap. Today every county negotiates alone, against counterparties who negotiate constantly, and the results range from genuinely transformative to nominal. A development intermediary that standardizes the benefit package, benchmarks it against project economics, and carries it from county to county converts scattered one-off deals into something closer to a market price for community consent — high enough to be worth defending locally, low enough that projects still clear.

SOURCES

Purdue Ag Economy Barometer via AFBF. Brookings; Data Center Dynamics, community-benefit agreements as standard practice. Power Magazine, Microsoft cost-recovery commitment; White House hyperscaler pledge.

A mission-driven development firm with a new economic model.

Community Compute brings together decades of experience at the intersection of AI, renewable energy, community permitting, and public communication — approaching development with a new set of principles and a new economic model. We begin with community solar and storage in the Northeast, where grids are constrained, state incentives are strong, and the first community-benefit frameworks are being written. Data-center development follows only once we hold community trust and a path that balances hyperscaler demand against long-term local value and environmental impact — air, sound, water, and energy.

Trust is the scarce asset in this market. We will not spend it early.

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